ChatGPT

Farm2Market AI Platform – UX Case Report

Farmer Experience Cases

1. Onboarding & Account Setup (Farmers)

Scenario: A Hsinchu farmer wants to join Farm2Market via web or LINE. The platform must make registration quick and simple, even for those with minimal tech experience.

- **Web Onboarding:** The farmer navigates to the Farm2Market web portal and selects **"Sign Up as Farmer."** They fill a short form (name, farm name, location, contact). The UI provides step-by-step guidance with clear labels and examples, so even a low-tech user can follow easily [1]. If needed, an **ID verification upload** is prompted (e.g. license or ID card) to establish credibility [2]. The system may also prompt to **link their LINE account** for notifications. Language selection is offered upfront or automatically set to Mandarin by location, with a toggle for English (more under Language Support).

- **LINE Chatbot Onboarding:** The farmer adds the Farm2Market LINE chatbot. On first contact, the chatbot greets in bilingual text and asks to choose a role: *"Are you a Farmer or Market Owner?"* (quick reply buttons). Upon choosing **Farmer**, the bot sequentially asks for essential info in a conversational manner (e.g. "What's your farm name?" then "What products do you grow?"). The chatbot uses **quick reply options and LIFF** mini-forms for data entry to keep interaction under 60 seconds. For example, after collecting basics, it might open a **LIFF webview** for uploading an ID or profile photo. Throughout, the process is **guided and broken into small steps**, ensuring even farmers with minimal smartphone experience can complete onboarding quickly [1]. Confirmation messages (e.g. "Your account is ready!") are shown in both Mandarin and English.

Smart UX: Both channels remember data once entered. If the farmer stops midway, the platform sends a **gentle reminder nudge** via LINE ("Hi! Complete your Farm2Market sign-up to start selling."). Defaults (like location via LINE GPS or previous inputs) are used to **autofill forms** when possible, minimizing typing for the user. The entire onboarding is optimized to finish in ~1 minute if the farmer has info handy.

2. Submitting Products (Listing Farm Produce)

Scenario: A farmer wants to list a new crop (e.g. organic tomatoes) on the platform with photos, description, and availability. This should be easy via both web and chat.

- **Web Interface:** After login, the farmer clicks **"Add Product."** A simple form appears with fields for product name, description, price per unit, quantity available, and optional tags (organic, non-GMO, etc.). The farmer can drag-and-drop or click to **upload photos** of the produce. The platform may offer **AI assistance**: for example, if a photo is uploaded, the system could recognize the produce type and suggest a title ("Tomatoes – Cherry Variety") or autofill some details. The UI might also show an **AI-generated quality grade** or freshness indicator as feedback (as seen in similar

platforms [3]). The farmer can toggle a switch to **"Translate description to English/Mandarin"**, upon which the system auto-generates a bilingual description [4]. They review and submit. The new product listing immediately becomes visible to local buyers via the marketplace.

- **LINE Chatbot:** The farmer sends a message like "Add product" or taps a menu option. The chatbot guides them through the listing in a Q&A style:

- **Photo:** "Please send a photo of the product." The farmer uploads a picture from their phone.
- **Name & Details:** The bot might reply with an image analysis: *"It looks like tomatoes. Please confirm the product name or edit:"* followed by a suggested name. The farmer can type or accept suggestion. Then the bot asks for a short description. If the farmer answers in Mandarin, the bot can offer to **provide an English translation automatically**.
- **Availability:** Next, it asks for quantity and available dates (with quick-pick options for dates or "available now").
- **Price:** The bot asks for price per unit. For ease, it could offer the last used price for this product as a quick button (form autofill).
- **Confirmation:** The chatbot summarizes: *"Tomatoes – 100 kg at $NT50/kg available now. Description: Fresh organic tomatoes… (with English translation). Confirm listing?"* The farmer confirms, and the product is posted.

The conversational flow is designed to be **concise (a few quick questions)** so that listing an item takes about a minute. Each step provides guidance or defaults (e.g., reusing last description or price) to minimize typing. On submission, the farmer receives a confirmation and the listing is instantly accessible to buyers nearby [3].

Smart UX: The platform might leverage AI to **enhance listings** – e.g., automatically tagging the product category, suggesting optimal pricing based on market data, or flagging if the photo is low quality (prompting the user to retake). If the farmer has listed this product before, the system can pre-fill details (description, past price) to speed up the process (the farmer just updates quantity). All instructions and confirmations are shown in the user's preferred language.

3. Receiving Demand Predictions & Reorder Reminders (AI Insights)

Scenario: The farmer wants to leverage the platform's AI to anticipate market demand and manage inventory. The system provides forecasts and timely reminders to assist the farmer's planning.

- **How it Works:** Farm2Market's AI analyzes aggregated data (market trends, buyer searches, past orders, seasonal patterns) to predict demand for various products [5]. For example, it might detect that local supermarkets are likely to need more leafy greens as a holiday approaches, or that the farmer's stock of rice is running low relative to typical buyer demand.

- **Web Dashboard:** On the farmer's dashboard, a **"Insights"** or **"AI Forecast"** panel displays key info:

- **Demand Forecast:** e.g., *"High demand expected for Sweet Corn next month (↑ 20%)*." This could be shown with simple graphs or color-coded alerts.
- **Pricing Trends:** e.g., *"Market price for organic tomatoes is trending up 5% this week."*

- **Reorder Reminders:** e.g., *"Store A usually reorders 50kg of rice every 2 weeks – next expected order in 3 days."* The interface might include a **"Reach Out Now"** button next to such reminders.

The farmer can click a suggestion to initiate action (e.g., prepare more stock or message the buyer).

- **LINE Notifications:** The AI proactively sends chatbot messages when it has useful predictions or reminders. For instance, Monday morning the farmer might get: *"📈 This week's demand forecast: High demand for cabbage in Hsinchu* [5] *. Consider harvesting an extra 20%.*" or " *Reminder: Your client GreenMart tends to reorder eggs around now. Ready to fulfill a new order?"*. Each message is brief and in the farmer's language, with options like **"View Details"** (opening a LIFF summary graph) or **"Contact Buyer"** to follow up directly.

These predictions and reminders help farmers make **data-driven decisions**, something key for optimizing their operations [5] . By receiving advance notice, farmers can adjust harvest or stock, reducing waste and avoiding missed sales. Reorder reminders act as nudges so low-tech users don't need to manually track buyer schedules – the system does it for them.

Smart UX: All insights are delivered in easy-to-understand terms (e.g., simple percent changes, icons for urgency). The system can be configured for the farmer's preference – e.g., daily or weekly summary of trends, and notifications only at appropriate times (not to overwhelm). Complex analytics are distilled into clear recommendations ("Plant more cabbage now!") rather than raw data. Crucially, the feature supports **bilingual output**, so a farmer can tap to see the information in English if needed (useful for learning or if sharing the data with an English-speaking partner). The automation ensures farmers stay informed **without having to navigate complicated charts**, which is friendly for low-tech users.

4. Generating Bilingual Marketing Materials

Scenario: The farmer wants to promote their farm and products to potential buyers. Farm2Market offers an AI tool to create marketing collateral (flyers, one-page product sheets) in both Mandarin and English, saving the farmer time and effort.

- **Web Flow (Marketing Studio):** The farmer accesses a **"Marketing Materials"** section. They choose a template (e.g., a flyer layout or a product info sheet). The platform auto-fills the template with the farmer's profile picture/logo, farm name, and a list of their products with images and descriptions. The AI then generates polished marketing text:
- A **farm introduction** (e.g., "Family-run organic farm in Hsinchu specializing in premium fruits").
- **Product descriptions** for each item, enhanced with selling points (seasonality, organic certification, etc.), translated professionally into the second language. For example, if the farmer entered details in Chinese, the system produces an English version side-by-side [4] .
- **Contact information** and a QR code (linking to their Farm2Market profile or LINE chatbot).

The farmer can edit any text (in either language) before finalizing. Once satisfied, they click **"Generate."** The result is a downloadable PDF or image flyer, formatted with bilingual text columns or sections. For instance, a one-pager might have Chinese text on the left and English on the right for each product, ensuring it's ready to share with any buyer.

- **LINE Chat Prompt:** A farmer can also trigger this via LINE by messaging something like "Create flyer" or selecting a " Marketing Kit" option. The chatbot would ask a couple of questions in chat for

quick setup: *"Which products do you want to include?"* (it may list their active listings with checkboxes in a LIFF view), and *"Target languages? (Chinese, English, or both)"*. After the farmer's selection, the AI generates the content. The chatbot then replies with a preview image or link to the full material. The farmer can tap the link to open the high-quality version or even directly share the image from LINE to a buyer group.

This feature helps farmers overcome the **pain point of marketing their goods** [6]. They can easily get a professional-looking flyer without design skills or hiring help. The bilingual aspect is crucial in Hsinchu, where local trade is in Mandarin but some communications (especially with corporations or export brokers) might require English.

Smart UX: The content generation uses templates and AI to ensure **consistency in branding and accuracy in translation**. For example, if a farmer calls a fruit "Dragon Fruit (火龍果)", the flyer will include both terms properly. The platform's AI references a library of agricultural marketing phrases to spice up descriptions (e.g. "crisp, sweet flavor" or "award-winning produce"), tailoring to each language's style. It even accounts for cultural nuances in phrasing. The UI provides **tooltips** explaining that the material will include both languages, so the farmer understands what will happen. By automating design and translation, this feature empowers even low-tech farmers to produce quality marketing content for broader reach [4].

5. Discovering Nearby Store Leads (Buyer Discovery)

Scenario: The farmer has produce to sell and wants to find new local markets or store owners to pitch to. The platform offers a way to discover potential B2B customers nearby, leveraging location data and platform buyer profiles.

- **Web Interface (Map & Directory):** Farmers can open a **"Find Buyers"** or **"Store Leads"** page. Here, they see a **map of Hsinchu and surrounding areas** with pins indicating registered market owners and stores. They can filter leads by type (e.g., grocery store, restaurant, supermarket) or by products the buyers are interested in. For example, a farmer selling organic vegetables might filter for "Organic markets" or see highlighted pins for stores that frequently purchase vegetables. Clicking a pin (or a list view entry) shows the store's profile: name, address, what products they seek, typical order size, and communication preferences. This interactive map makes it intuitive to find local leads [3], much like how buyers can find nearby farms.

- **LINE Chatbot:** The farmer can type "Find nearby stores" or share their location via LINE. The chatbot responds with a carousel of **top nearby leads**:

- Each carousel card could display a store's name, a short description (e.g., "Local organic grocery, 5km away"), and buttons like **"View Profile"** or **"Contact"**. Swiping shows multiple stores.
- If the farmer clicks "View Profile," a LIFF page may open with full details and past interactions if any.
- The chatbot might also directly say: *"I found 3 nearby businesses looking for produce: GreenMart (2km), Hsinchu Fresh Market (5km), Local Restaurant (3km). Which would you like to contact?"* with quick reply options.

By integrating map and search functionalities, the platform proactively surfaces **relevant leads** instead of making farmers do cold outreach blindly. It leverages the platform's network data: for instance, highlighting a new supermarket that just signed up in the area, or a restaurant known to buy in bulk. This is especially useful in bridging the gap for farmers who don't traditionally network with store owners.

Smart UX: The lead suggestions can be **personalized by AI**. If the system knows a farmer grows a lot of mangoes, it will highlight buyers who often purchase mangoes. It might even show a "match score" or a note like *"This store's demand matches your supply 90%."* For low-tech users, the interface remains straightforward: visual map pins or simple list entries with big buttons. The design goal is that within a 60-second glance, the farmer can identify a new lead and initiate contact. Additionally, the **language toggle** applies here too – a store profile might be originally in Chinese, but an English toggle will show translated info if the farmer prefers. This ensures no language barrier in understanding what a store needs.

6. Outreach Automation & Negotiation Handling

Scenario: After identifying a potential buyer, the farmer wants to reach out and negotiate a deal. The platform assists by automating initial outreach messages and guiding the negotiation process.

- **Automated Outreach:** On a store's profile (web or via chatbot), the farmer clicks **"Contact Store"** or **"Send Introduction."** The platform provides a pre-written message template, customizable before sending. For example, it might draft: *"Hello, I'm a local farmer from Hsinchu offering fresh organic tomatoes and lettuce. I saw your store profile and think we could work together. Interested in a sample or price list? (Message sent via Farm2Market)"* – in both languages if needed. The farmer can simply hit send, or edit details (the AI ensures translations stay aligned). This saves farmers who are not comfortable with formal outreach the trouble of composing messages.

- **Conversational Interface:** If the buyer responds, they can chat through the platform. **Web:** A messaging interface akin to an email/chat thread is available on the dashboard (with notifications for new messages). **LINE:** The conversation can happen directly via the chatbot, which relays messages between the farmer and buyer (potentially with auto-translation if they prefer different languages). The platform encourages **direct communication** to build trust [7], but with AI support in the background.

- **Negotiation Assistance:** As price and terms are discussed, the AI assistant can chip in subtle guidance (optionally visible to the farmer as tips). For example, if a buyer says, "Can you offer 100kg at a lower price?", the system might highlight a **suggestion tip** only the farmer sees: *"Suggested response: Offer 100kg at 5% discount if they commit to monthly orders"* or *"Your minimum acceptable price based on costs: NT$40/kg."* This helps less experienced farmers negotiate confidently. The AI can draw on market price data and the farmer's own cost inputs to advise on a fair deal.

- **Time-Saving Automation:** The platform can automate routine parts of negotiation. For instance:

- If a buyer asks for the farmer's **product catalog or price list**, the AI can immediately compile and send a neatly formatted list of the farmer's current products and prices.
- Scheduling: If during chat the buyer says, "Can we talk on Friday?", the AI can detect this and prompt a meeting scheduler.
- Language gap: If the buyer writes in English and the farmer is more comfortable in Chinese, the chatbot interface could automatically show the farmer a Chinese translation and allow them to reply in Chinese, which is then sent as English to the buyer. This seamless translation fosters smooth negotiation without embarrassment.

Smart UX: For low-tech users, the negotiation UI avoids overload. Instead of complex contract forms, it uses familiar chat paradigms with enhancements. The platform might include **quick-reply buttons** in chat like "Accept offer", "Propose new price", or "Add delivery details" to structure the negotiation. By clicking these, the farmer triggers automated messages or form pop-ups rather than typing everything. This ensures the negotiation doesn't stall due to uncertainty – the system guides them on **next steps**. All communication remains logged in one place for reference, and either party can escalate to a voice call if needed (the platform can help schedule it). Overall, the system's direct communication features strengthen farmer-buyer relationships while reducing misunderstandings [7].

7. Follow-Up & Performance Tracking

Scenario: After closing deals, farmers need to maintain relationships and evaluate their business performance. Farm2Market provides tools for follow-up and analytics tracking.

- **Post-Deal Follow-Up:** Once an order is delivered, the platform might prompt both parties for feedback. The farmer's dashboard (and LINE bot) notifies: *"Order #1001 delivered to GreenMart. Record feedback?"*. The farmer can input a short note (e.g. "Delivery successful, quality good") for their records. The buyer might rate the produce or provide comments via their interface. Positive reviews show up on the farmer's profile – building credibility for future buyers [8]. If the buyer hasn't reordered in a while, the system might nudge the farmer: *"It's been 30 days since GreenMart's last order. Consider checking in."* with a one-tap option to send a prepared "just checking" message.

- **Performance Dashboard:** On web, farmers have a **"Statistics"** or **"Performance"** page updated in real-time [9]. Key metrics are displayed with simple visuals:

- **Sales Volume:** Total and per product (e.g., "Tomatoes: 500kg sold this month").
- **Revenue Earned:** Perhaps in NT$ and a trend compared to last month.
- **New vs. Repeat Buyers:** How many new market contacts were made, and how many repeat orders.
- **Fulfillment Stats:** On-time delivery rate, response time to inquiries, etc.
- **Customer Feedback:** Average rating or a list of recent testimonial quotes.

Low-tech users benefit from this being not just raw numbers but also plain-language insights: e.g., *"Your best-selling product this quarter is Green Onions"* or *"You gained 3 new buyer connections this month."* Graphs are kept simple or optionally detailed.

- **LINE Summary:** The chatbot can provide performance info on-demand or periodically. If the farmer types "How am I doing?" or at month-end, the bot replies with a brief summary: *"This month: 3 orders, 2 new buyers, 1000kg sold. Top product: Cabbage. ⭐ Average buyer rating: 4.5/5. Great job!"* For more details, it might offer a link to the web dashboard or a CSV report download if the user wants that.

- **Continual Learning:** The platform might also include a **"Tips"** section in follow-up. For example, if the data shows slower sales of a particular crop, it could suggest: *"Consider creating a promotion for carrots, as sales dipped 10%. Click here to generate a promo flyer."* These proactive tips drive usage of platform features (like marketing material generation, etc.) and help farmers improve.

Smart UX: This tracking addresses a crucial need: helping farmers see the **big picture of their business** [9]. By automating data collection (the farmer doesn't have to manually track sales in a ledger), it reduces their

workload. The information is presented in **clear language with minimal jargon**, and bilingual labels if needed. The design uses icons and color-coding (green up arrows for growth, red down arrows for declines) to be glanceable. The follow-up reminders and tips act as a virtual assistant, ensuring the farmer stays engaged with buyers (e.g., sending a thank-you note or a re-engagement offer) without having to remember all these tasks themselves. Ultimately, the platform fosters continuous improvement by making performance data accessible and actionable to farmers in an intuitive way.

Market Owner Experience Cases

1. Initial Signup & Role Selection (Market Owners)

Scenario: A market owner (e.g., a grocery store manager in Hsinchu) hears about Farm2Market and wants to sign up to find local farm produce. The onboarding should quickly identify their role and get them to relevant features.

- **Web Signup:** The market owner goes to the Farm2Market site and begins registration. They choose **"Sign Up as Buyer/Market"** (the terminology on the button is clear). The form asks for name, business name, type of business (dropdown: retail, restaurant, etc.), business address (for locating nearby farms), and basic contact. They might also need to provide a business license number or VAT (optional for verification). Similar to farmers, the onboarding is streamlined and guided. If the owner accidentally started as a farmer, they can switch role easily. Once submitted, they can immediately browse farm listings. The UI even asks if they want to **"find local produce now"** and offers a quick tutorial of the interface.

- **LINE Onboarding:** The market owner adds the official LINE chatbot. The bot's first question, *"Are you a Farmer or Market Owner?"*, when answered with **Market Owner**, triggers a tailored sign-up flow. The chatbot collects minimal info: "Please send your name or store name", "What kind of products are you looking for?" (this could be multiple choice: fruits, veggies, dairy, etc. to personalize later suggestions), and possibly asks to share location via LINE (for finding nearby farms). The bot may also prompt an optional LIFF form if more details are needed (like entering business address on a map). Within a few steps, the account is created. The chatbot confirms *"Welcome, you're signed up as a Market Owner!"* and immediately perhaps offers a starting point like **"Browse farms near you"** with a quick button.

This initial setup ensures the platform knows the user's role and preferences, directing them to the relevant UX flows. The emphasis is on **ease and speed**: providing just enough info to get personalized results without a long registration. As with farmers, all text is bilingual and user-friendly.

Smart UX: Role selection is critical – the system clearly **delineates features for farmers vs. market owners** to avoid confusion. The onboarding flow might use icons or illustrations (on web) to depict a farmer vs. a buyer persona when choosing role. Also, to assist low-tech business owners, the sign-up refrains from heavy paperwork at first. Verification of business (if needed for large orders or payments) can be deferred or made optional. The platform could integrate LINE Login such that a user's LINE display name is imported to speed things up. And like with farmers, a partially completed signup triggers reminders, ensuring busy store managers can pick up where they left off.

2. Browsing Local Farm Listings

Scenario: The market owner wants to find produce available from nearby farmers. The platform provides an intuitive browsing experience to discover and evaluate farm listings.

- **Web Browsing:** Upon login, the market owner lands on a **Marketplace Home**. This page showcases local farm produce in a way that's easy to scan:
- A **search bar** (with suggestions) to find a specific product (e.g., "strawberries").
- **Filter options**: by category (vegetables, fruits, grains), by distance (e.g., within 20 km), by farming practice (organic, GAP certified), etc. [10] .
- A dynamic **map view** with pins for farms, alongside a list view. The owner can toggle map/list. The map shows the location of farms; clicking a pin highlights that farm's listings. This was effectively used in similar platforms to help buyers locate products in real time [3] .
- Each listing card displays the product name, a thumbnail image, price, available quantity, and the farm name. For example: *"Organic Tomatoes – NT$50/kg – 100kg available – Green Valley Farm (5 km away)"*. If the market owner set preferences at signup, the homepage might prioritize those (e.g., show fruit listings first if they indicated interest in fruits).

The market owner can click a product to see a **detailed listing page**: larger photos, full description (bilingual), farmer's profile info (ratings, other products), and any AI quality grade or notes (like "A-grade freshness" if provided). From here they can take actions: **"Request Sample"**, **"Message Farmer"**, or **"Add to Order."**

- **LINE Browsing:** The chatbot can facilitate discovery in a conversational way:
- The owner might type "Find apples" or use a menu like "Browse by category". The bot could respond with a list or carousel: *"Found 5 farms with apples:"* each with a small image and name/price. The owner can tap one to get details.
- Alternatively, the bot might support a **guided browse**: *"What are you looking for today? Fruits / Vegetables / Others"* (quick buttons). If "Vegetables", it might ask "Any specific one?" or just list popular ones.
- The chatbot can also send a **daily highlight**: *"New produce near you: Farmer Lee has fresh bamboo shoots."* which encourages discovery.
- For each item, a **"View Details"** button can open a LIFF mini-app with the detailed listing page, since showing lots of info in chat is cumbersome. That page would be mobile-optimized and similarly bilingual.

Browsing is designed to be accomplished in under a minute: quick filters to narrow down, visual cues like farm distance, and straightforward navigation. The result is that a market owner can **quickly locate the produce they need** without hassle, addressing the common pain of spending too much time finding reliable suppliers.

Smart UX: Recognizing that market owners might be busy or on-the-go, the interface could include a **"smart search"** function: as they type or select, the system could auto-complete queries (e.g., type "lett" suggests "lettuce") and show result counts. The map vs. list view caters to different preferences – visual/spatial or textual list. The platform might also personalize the feed: e.g., if a user frequently buys cabbage, new cabbage listings might be highlighted ("New Supplier in Town!"). **Multi-language support** means if a listing is entered in Chinese, an English-speaking buyer can still search in English ("cabbage") and find it, as the

system tags it appropriately and can display the English description [4]. All these reduce friction, making the browsing experience efficient and user-friendly [10].

3. Requesting Product Samples

Scenario: Before committing to a new supplier, a market owner may want to try the product. The platform streamlines the process of requesting a sample from a farmer.

- **Web Workflow:** On a product detail or farm profile page, the market owner clicks **"Request Sample."** A dialog or form pops up asking for a few details:
- **Sample Quantity:** Defaulted to a reasonable small amount (e.g., "5 kg" or "1 crate", depending on product) which the owner can adjust.
- **Delivery/Pickup Preference:** Options might include "Deliver to my store (address)" or "Visit farm for pickup". If delivery, the owner's address (from profile) is auto-filled [4]; if pickup, perhaps the farm address and an option to schedule a visit.
- **Preferred Date/Time:** e.g., "As soon as possible" or select a date/time slot. The system might show the farmer's available visiting hours or estimate a delivery date.
- **Comments:** A text box for any special request (e.g., *"Please ensure the apples are the Honeycrisp variety"*).

The owner submits the request. The UI then shows the request status (pending acceptance by farmer). The farmer is immediately notified.

- **LINE Chatbot Workflow:** The owner can type "request sample of [product]" or, after viewing a listing via chatbot, tap a **"Request Sample"** quick button. The bot will then walk through the necessary info:
- Confirm product and quantity: *"Request a sample of Tomatoes. Default amount is 2kg. Change amount or OK?"* (The user can enter a different quantity or just "OK".)
- Delivery or pickup: *"Would you like it delivered or to pick up at the farm?"* (two buttons). If delivered, the bot confirms the address: *"Deliver to Hsinchu Main Store, 123 Zhongzheng Rd?"* (address on file) – user can confirm or provide an alternate. If pickup, it may reply with the farm's location and ask *"When would you like to visit?"*
- Schedule: The bot might offer a couple of suggested time slots (synced with a calendar or the farmer's set availability) or let the user type a preferred time.
- Extra note: *"Any note to the farmer?"* The user can skip or add a short message.

The chatbot then confirms: *"Sample request sent to Green Valley Farm for 2kg Tomatoes, delivery to your store by tomorrow."* The entire interaction is designed to be completed in a minute or less, with minimal typing due to intelligent defaults.

Once the request is sent, the farmer will accept or propose changes. The market owner will receive an update (e.g., *"Farmer Lee accepted your sample request. Expect delivery by Friday."*). If via LINE, this comes as a message; via web, as a notification or email.

Smart UX: Automating the sample request saves time compared to phone calls or emails. The platform knows the context (which product, who the farmer is, who the buyer is) so it auto-fills wherever possible (address, product info) [4]. It also ensures nothing is forgotten (date, amount) by structuring the input. For low-tech users, this structured approach reduces errors – they won't accidentally forget to tell the farmer where to send the sample, for example. Additionally, the system could integrate with a **logistics partner** for

sample deliveries: if the farmer opts in, the platform could generate a shipping label and tracking for the sample. The market owner would then be able to track the sample's delivery just like a normal order. This level of service makes the platform feel full-service and **builds trust** early in the relationship.

4. Scheduling Calls or Meetings

Scenario: A market owner is interested in a farm's products and wants a direct conversation or a farm visit. The platform provides a convenient scheduler to set up calls or meetings between buyers and farmers.

- **Web Scheduling:** On the farmer's profile or after some chat exchanges, the market owner clicks **"Schedule a Call/Meeting."** This brings up a scheduling interface:
- If it's a call, the platform might show the farmer's available time slots for a phone or video call (if the farmer has synced their availability or set a schedule). The owner picks a slot and confirms. The system sends calendar invites to both parties (and possibly a link if it's a video call).
- If it's a farm visit, similarly, available visiting slots or a date picker appears. The owner selects a date/time and the farm address is shown for confirmation. They confirm the meeting.
- The interface auto-fills details like time zone and contact method. It might allow the owner to choose **"LINE call"**, **"Phone call"**, or **"Zoom meeting"** as options for how to connect.
- After scheduling, the appointment appears in a **Meetings** section of the dashboard with details. Reminders are sent 1 day and 1 hour before the meeting.
- **LINE Scheduling:** The owner can simply message the chatbot, *"Arrange a call with Green Valley Farm"*, or use a quick command after some chat like " Schedule Meeting". The bot will then interact:
- *"Great, you'd like to talk to Green Valley Farm. Would you prefer a phone call or an in-person visit?"* (User chooses).
- If phone call: *"Here are some times Farmer Lee is available:"* followed by a few options (e.g., "Tomorrow 3pm", "Friday 10am") as quick reply buttons, derived from the farmer's calendar settings. The owner selects one.
- If visit: *"Farmer is available on-site on Tue and Thu mornings. Pick a date:"* and then *"What time roughly?"* Possibly it uses a date/time picker in LIFF for more precision.
- Once a choice is made, the bot confirms: *"Meeting scheduled on Thu 10am at Green Valley Farm (address: …). A calendar reminder will be sent to you."* If a call, *"Call scheduled on Fri 3pm. You will receive a LINE call from the farmer."*

In both channels, the goal is to make setting up meetings **as easy as booking a calendar appointment**, without back-and-forth emailing. By offering suggested slots and using the platform's knowledge of both parties' preferences, it cuts down negotiation time to under a minute.

Smart UX: The platform's smart scheduling avoids common pitfalls: - It can automatically handle **time conflicts** – if a chosen slot is taken, it suggests alternatives. - It sends **reminders and updates** automatically (via LINE message or email). - It respects language: the invitation details might be bilingual (so both parties are clear). - For low-tech users, it integrates with what they know – e.g., LINE call for those who prefer not to dial phone numbers, or just exchanging contact if needed. - The system also updates status if plans change: e.g., if farmer proposes a new time, the bot can ask the owner to accept or suggest another, structuring the reschedule process. - All of this ensures even someone not familiar with digital calendars can successfully arrange a meeting through guided steps and clear notifications.

5. Confirming Orders & Issuing Purchase Orders (POs)

Scenario: After discussions or sample approval, the market owner is ready to place an order for products. The platform facilitates order confirmation and generates a formal Purchase Order for record-keeping.

- **Placing an Order (Web):** The market owner adds desired items to an **order cart**. This could happen from browsing (e.g., clicking "Add to Order" on a listing) or directly from a negotiated deal (the farmer might send a quote that the owner can accept). On the **Order Review** screen, they see a summary of items (e.g., *Tomatoes – 50kg @ NT$50/kg = NT$2,500*). The owner can adjust quantities before finalizing. If everything looks good, they click **"Confirm Order."**

At this point, the platform might prompt for any **additional terms**: delivery date, special instructions (like "need refrigerated transport"), etc., if not already specified. The owner confirms these details. The platform then **generates a Purchase Order document** with a unique PO number, listing buyer and seller information, items, prices, total amount, and agreed delivery date. This PO is essentially a formal agreement.

The UI shows "Order Confirmed!" and provides a link to download the PO (PDF) or view it. A copy is emailed to the market owner and to the farmer for their records. The order now appears in both parties' **Orders section**, with status "Pending Delivery."

- **LINE Order Confirmation:** If the negotiation happened via the chatbot, the flow is slightly different but still streamlined. Suppose the farmer says: *"I can supply 50kg at NT$50/kg next week"* and the owner replies *"Okay, let's confirm that."* The bot can detect this intent to confirm order. It then summarizes: *"You are ordering 50kg Tomatoes from Green Valley Farm at NT$50/kg, delivery by next week. Total NT$2,500. Type 'yes' to confirm or 'edit' to adjust."*

If the user types **"yes"**, the bot responds with an order confirmation message: *" Order confirmed! Your Order #1001 is placed."* It then provides a button or link: **"View Purchase Order"** which opens a LIFF page to display the PO details (in a nicely formatted manner, possibly bilingual). The owner can save or share this document if needed.

If the user typed "edit" or needed to change something, the bot would prompt through adjustments similarly to a form (though major edits might be easier on web). But ideally, by the time they confirm via chat, the details are already agreed upon in conversation.

In either channel, the emphasis is on making the confirmation **quick and clear**. The platform ensures that all important details are captured so the order is binding and understood by both sides.

Smart UX: To assist low-tech users and provide guidance: - The platform can pre-populate standard terms on the PO (e.g., payment terms, delivery terms) to avoid confusion. It might use templates that are common in the industry. - The fact that a formal PO is generated adds professionalism, which market owners expect. It's in **both English and Mandarin**, so that it can be used for any internal process or even for cross-border deals. - The system logs everything, so if the owner later forgets what was agreed, they can easily retrieve the order info. - Additionally, if the platform supports **integrated payments**, at confirmation it might also offer a payment option (like "Pay Now" to pay online, or instructions for bank transfer). However, even without immediate payment, the key is the order record is created. - The UI keeps the process within the

"≤60 seconds" philosophy by having all info ready to review in one screen or message before confirming, avoiding lengthy forms. The user basically just reviews and hits confirm.

6. Post-Sale Communication & Delivery Tracking

Scenario: After an order is confirmed, the market owner needs to track fulfillment – ensuring the produce is delivered on time – and communicate with the farmer for any post-sale needs (issues, feedback, repeat orders).

- **Delivery Tracking:** Once the farmer ships or dispatches the order, they update the order status (via their app or chatbot, e.g., marking as "Shipped" and possibly entering a tracking number or delivery agent details). The market owner can then:
- **Web:** Check the **Orders** page where each order has a status timeline (e.g., Ordered → In Transit → Delivered). If a tracking number is provided (with a linked courier), a **"Track Package"** button opens the courier's tracking page or shows the transit status right there [11] [12]. If the platform has integrated logistics partners, a map might show the delivery vehicle en route, or at least an ETA.

- **LINE:** Receive push updates: *"Your order #1001 has been shipped and is expected by Oct 25."* Later: *"Order #1001 delivered. Please confirm receipt."* The owner can type "Received" or just tap a confirm button in chat, which finalizes the order. If something's delayed, the chatbot might proactively inform: *"Order #1001 is delayed, now arriving Oct 26 – sorry for the inconvenience."*

- **Post-Sale Communication:** If the market owner has an issue (say the quality is not as expected or quantity mismatch), they can easily contact the farmer through the same communication channel:

- **Web:** On the order details page, there's a **"Contact Seller"** or **"Report Issue"** button. Clicking it opens a chat or form to describe the issue. The platform encourages direct resolution first, maybe opening a chat window linked to that order for context.

- **LINE:** The owner can just send a message in the existing chat thread, e.g., "Noted the lettuce was slightly wilted." The farmer can respond, and if needed, they work out a compensation or solution. The platform can escalate to support if not resolved, but at least it provides a seamless way to talk post-sale.

- **Confirming Delivery & Feedback:** After marking an order as received, the market owner might be prompted to rate or give feedback: *"How was the quality of the produce from Green Valley Farm? [☆ 1–5]"*. They can select a star rating and add a comment. This feedback is sent to the farmer (and possibly displayed on their profile as part of reviews). It closes the loop on that transaction.

- **Reorder Convenience:** If the owner is happy and wants to reorder the same items, the platform makes it easy:

- On web, an **"Reorder"** button on past orders can recreate a new order with same items (the owner can adjust quantities if needed, then confirm).
- On LINE, the bot might remind: *"It's been 2 weeks since your last tomato order. Would you like to reorder the same amount?"* If the owner says yes, the bot triggers a new order flow with minimal steps.

Smart UX: The post-sale phase is critical for satisfaction. The platform's design ensures transparency and trust: - **Real-time tracking** provides peace of mind to the buyer, much like consumer e-commerce deliveries [11]. Seeing updates reduces the need to manually call the farmer for "where is my order?". - By keeping communication lines open (through chat or on-platform messaging), any issues can be quickly addressed – which is crucial when dealing with perishable goods. The system's **24/7 support** aspect (if any) might also be accessible here, but often direct farmer-buyer chat suffices [13]. - The user doesn't have to navigate away or use external tools; everything from order to delivery to feedback is in one integrated flow. - All interactions continue to be bilingual as needed: e.g., a tracking update can be shown in the user's language, and if the farmer writes a message in Chinese, an English-preferring buyer could get a translated version by toggling. - These thoughtful touches like reorder suggestions and quick issue resolution forms demonstrate **user guidance and automation** – the platform nudges the buyer at the right times (delivery received? rate it. Time to reorder? one-click to do so) so that even a busy store owner doesn't miss important actions. It keeps them engaged and builds loyalty through convenience and reliability.

Cross-Channel UX Coverage: Web & LINE Chatbot Integration

Farm2Market is designed as a **multi-channel platform**, accessible via a traditional web interface and a LINE-native chatbot (via LIFF). The UX flows described above are crafted to work seamlessly on both, ensuring users can accomplish tasks on their preferred channel without compromise. Key considerations for cross-channel design:

- **Consistent Core Experience:** Whether on web or LINE, users have access to all main features (onboarding, listing, browsing, messaging, etc.). The interface may differ (graphical vs. conversational), but the outcome is the same. For instance, a market owner can confirm an order through a web form or simply by replying "yes" to the chatbot's summary – both result in a confirmed order with a PO generated.

- **Optimized for Speed (<60s interactions):** On web, tasks are streamlined with intuitive UI elements (clear buttons, short forms) to be completed quickly. On LINE, tasks are broken into bite-sized chat interactions with quick replies and defaults to avoid long typing. The design goal is that common actions (posting a product, finding a lead, placing an order) take around a minute or less to execute, keeping users engaged and not overwhelmed. This aligns with best practices for mobile and chat UX, where users expect fast, snackable interactions.

- **LIFF for Complex Tasks:** The integration of **LINE Front-end Framework (LIFF)** is used to handle any workflows that benefit from a richer UI than chat bubbles can offer. For example, viewing a full product catalog, filling in a detailed form (like scheduling with a calendar picker), or viewing a formatted document (like a Purchase Order or a flyer) can all open within LINE as a mini-webpage. This approach provides the flexibility of web UI within the LINE app. The transition is smooth: the user taps a link or button in chat, completes or views the content in the LIFF view, and then returns to the chat when done. It ensures even detailed tasks don't become tedious in a chat format.

- **Notifications and Continuity:** Actions taken on one channel reflect on the other in real time. A farmer might upload a product via the web and immediately the LINE bot can send them a confirmation message. A market owner could start a conversation with a farmer on the web

messaging interface and continue it on LINE on their phone. This continuity means users can switch channels as convenient (e.g., a farmer in the field might use LINE voice input to add a product, then later use the web dashboard at home to analyze performance).

- **Channel-Specific UI Elements:** The design leverages the strengths of each platform:

- On **Web**: richer data display (maps, tables, charts), multi-column layouts for desktops, and possibly more detailed analytics. Good for when users have time to sit down and manage a lot of information.

- On **LINE**: instant communication, push notifications, and quick actions. Great for immediate needs and updates (e.g., getting an alert of new demand, quickly responding "yes/no" to a question). LINE's ubiquity among Taiwanese users means the chatbot can serve as a friendly assistant that users interact with like they would with a person, which lowers the learning curve.

- **User Guidance on Each Channel:** Both channels incorporate guidance but in different ways. On web, there might be tooltips, highlighted tour steps for new users, and well-labeled buttons. On LINE, guidance comes as the bot's prompts and suggestions in conversation. In both cases, the tone is encouraging and helpful, tailored to users who may not be tech-savvy. The underlying principle is to **make the tech feel invisible** – the user should feel like they're just naturally accomplishing their task, whether clicking a button or answering a quick question in chat.

By covering both web and LINE, Farm2Market meets users where they are. Farmers working in fields might find it easiest to use their phone and LINE app to interact, whereas a market owner in an office might prefer a desktop for browsing lots of product listings. The platform ensures neither experience is siloed – they are two doors to the same marketplace, both optimized for speed, simplicity, and reliability.

Language Support: Mandarin & English Throughout

Language support is a first-class feature of Farm2Market's UX design. All user flows, interfaces, and generated content are available in both **Mandarin Chinese** and **English**, reflecting the bilingual needs of Taiwan's business environment and potential international expansion. Key aspects of language support include:

- **Bilingual Interface Text:** Every label, button, and prompt on the web platform can be toggled between Chinese and English. Users can select their preferred language at onboarding or switch at any time via a visible language toggle (e.g., a globe icon or "中/En" switch). The LINE chatbot similarly detects the user's language – if the user's LINE app is in Traditional Chinese or if they initially speak Chinese to the bot, it will reply in Chinese. It can seamlessly switch if the user switches languages. This ensures that whether a farmer is more comfortable in Mandarin or a buyer prefers English, they can use the system with ease [14].

- **Multilingual Content Creation:** As noted in the farmer use cases, when a user inputs content (product descriptions, messages, etc.), the platform can provide instant translations. Farmers writing descriptions in Mandarin can get an English version generated for display to English-speaking buyers, and vice versa [4]. This is crucial for things like the **marketing materials** generator and even chat – e.g., a buyer can write in English and the farmer sees Chinese, powered by behind-the-

scenes translation AI. The platform effectively acts as an interpreter, which broadens the reach of local farmers to international or non-Chinese-speaking clients.

- **System Messages and Notifications:** All automated messages (like demand forecasts, order updates, reminders) are crafted to be bilingual or in the user's chosen language. In some cases, the message might be combined (Chinese sentence followed by English sentence) to avoid any ambiguity, especially for critical notifications. However, most of the time it will stick to the user's language preference. The goal is to never present a user with text they cannot read. Even the **Purchase Orders and documents** generated are bilingual formatted, so they serve as official documents understood by all parties without needing translation.

- **Localizations and Units:** Beyond language, the UX accounts for local conventions. Since it's Hsinchu, Taiwan-based, the default language is Traditional Chinese and units like weights (catty vs. kg) or currency (NT$) are set accordingly. If switched to English, units remain the same (since NT$ and metric are used, likely consistent), but the formatting of dates or numbers might adapt to international style if needed. All of this falls under comprehensive localization support [14] to make the app feel native to different users.

- **Help and Support in Both Languages:** Any help center articles, FAQs, or customer support chat would also be bilingual. For example, a low-tech farmer might call a support line – they should have Mandarin service. Within the app, if they type "help" in Chinese, they get Chinese guidance. Ensuring support materials in both languages is essential for user adoption.

By offering robust bilingual support, Farm2Market not only caters to the Taiwanese bilingual environment but also **lowers barriers** for users. Farmers who only speak Mandarin can still connect with an English-speaking business owner through the platform without language being a bottleneck. This inclusion is a key feature noted in successful agriculture platforms, as multi-language access broadens the user base and comfort [15] [16]. It demonstrates cultural sensitivity and practical utility, increasing the platform's value for all users.

Smart UX Recommendations for Low-Tech Users

To ensure the platform is accessible and friendly to users who may not be tech-savvy (which is common among traditional farmers and some smaller market operators), Farm2Market incorporates **smart UX features** that automate tasks and guide users gently through the workflows:

- **Step-by-Step Guidance & Wizards:** As highlighted in onboarding, every complex task is broken into simple steps or wizard-like flows [17]. The interface avoids overwhelming the user with too many fields or options at once. For example, listing a product via chatbot comes one question at a time, and on web, the form might be split into sections with a progress indicator. This guides users through completion without confusion.

- **Autofill and Remembering Preferences:** The system remembers user data to minimize repeat typing. If a farmer has entered their address or standard package weight once, the next time the system can auto-populate that info. When a market owner requests a sample, their address is already filled in [4]; when a farmer creates a new listing, the description template might pre-load

with last used wording if it's similar produce. This not only saves time but also reduces errors (a key usability improvement for those less comfortable with typing).

- **AI-Powered Suggestions:** The platform uses AI not just for analytics but also for micro-copy suggestions. For instance:

- When writing a product description, a farmer might see a prompt *"Need help? Auto-generate description"*. Clicking it uses AI to draft a nice description which they can accept or edit.
- When negotiating, as discussed, suggestion chips might appear to help the user respond. Similarly, if a user is scheduling a meeting, the system might highlight *"mornings are usually available for the farmer"* to guide their choice.

- If a farmer hasn't uploaded a profile picture or farm description, the system could nudge them with *"Profiles with a picture get 3x more buyer interest* [18] *. Add one now?"* and a one-click action to do so.

- **Reminders & Notifications:** To compensate for users who might forget to check the app, important actions trigger reminders:

- If a farmer doesn't respond to a new message or order request within a few hours, the LINE bot pings them: *"You have a new buyer inquiry, please respond to maintain a good response rate."*
- If a market owner has an unpaid invoice or hasn't confirmed delivery, the system reminds them to do so to close the loop.

- These reminders are phrased helpfully, not nagging, and can be customized (e.g., a farmer can choose to get a summary at end of day rather than instant notifications if they prefer).

- **Error Prevention and Clarification:** The UX design anticipates common mistakes of low-tech users and tries to prevent them. For example:

- If a farmer enters an unusually high price by mistake (say NT$5000 instead of NT$500), the system might prompt *"Did you mean 500? (add one less zero)"* before finalizing.
- If a market owner tries to order more than what's listed as available, the UI clearly shows the max and prevents over-selection.

- Tooltips or short helper texts accompany complex concepts – e.g., explaining what a "Purchase Order" is in simple terms the first time, or what "GAP certification" means if it appears.

- **Visual Aids and Simplicity:** Icons, images, and visual cues are used wherever possible instead of text. For instance, a farmer's menu might use icons: 📦 for Products, for Alerts, 📊 for Stats – coupled with text labels. This helps those who may not read long text comfortably. The design uses **clean layouts and larger touch targets** on mobile knowing that precision may be difficult for some users.

- **Demo and Support:** The platform may include a quick demo mode or sample data so new users can explore without fear. Also, a clearly accessible **Help** section (with Q&A and possibly video tutorials) is provided. Low-tech users often benefit from short videos or illustrated guides (which could be

delivered via LINE as well, e.g., “Watch how to add a product” 30-second clip). And if they get stuck, a support chatbot or hotline is one tap away, providing human assistance in their language.

By embedding these smart UX elements, Farm2Market ensures that even users unfamiliar with digital marketplaces can onboard and use the platform effectively. The approach is empathetic: understanding user pain points (as gleaned from research like difficulty marketing goods, or not knowing how to reach buyers [6]) and directly addressing them with feature solutions. This thoughtful design lowers the learning curve and builds user confidence, making the platform truly inclusive for the farming community and small business owners.

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